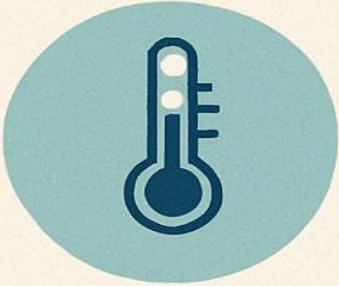
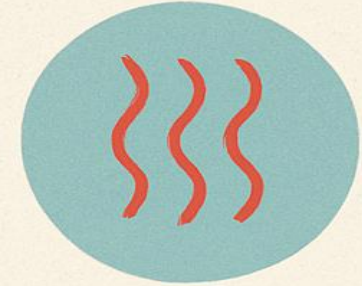


THYROID DISORDERS



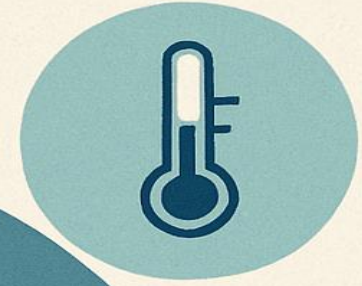
FATIGUE



HAIR LOSS



WEIGHT GAIN



HEART RATE



Hypothyroidism

Hypothyroidism refers to a deficiency of thyroid hormone in the adult, resulting in slowed body metabolism due to decreased oxygen consumption by the tissues and pronounced personality changes.

Myxedema is a complication of hypothyroidism characterized by a generalized hypo-metabolic state. Myxedema coma is a life-threatening situation in which all body systems are severely compromised by the hypo-metabolic state.

Etiology: *There are several types of hypothyroidism:*

Primary

```
graph TD; A[Primary] --> B[Secondary]; B --> C[Tertiary];
```

Secondary

Tertiary

Primary hypothyroidism may be caused by

**Congenital
defects of
Thyroid**

**Defective
hormone
synthesis**

Iodine deficiency

**Surgery or
radioactive
therapy for
hyperthyroidism**

**Antithyroid
drugs**

**Following
chronic
inflammatory
diseases**

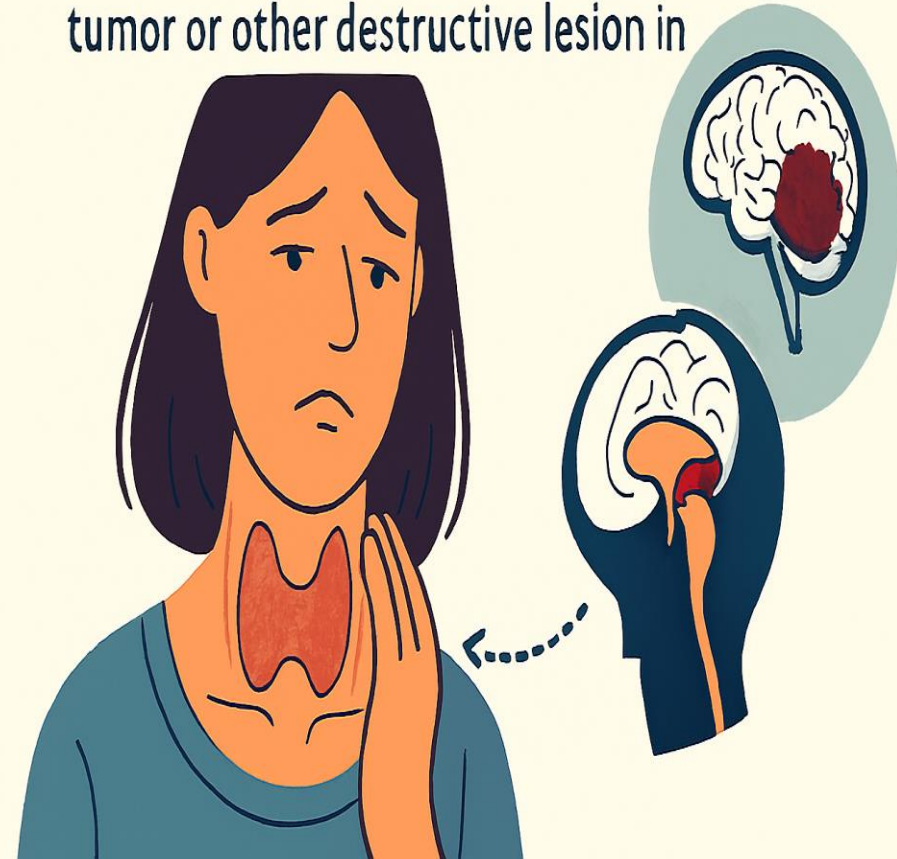
SECONDARY HYPOTHYROIDISM

Develops when there is insufficient stimulation of the normal thyroid gland



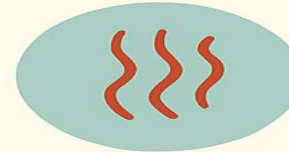
TERTIARY HYPOTHYROIDISM

Can develop if the hypothalamus fails to produce thyroid-releasing hormone (TRH). This may be due to a tumor or other destructive lesion in



CLINICAL MANIFESTATIONS

1. Extreme fatigue
2. Hair loss, brittle nails and dry skin
3. The voice may become husky and the patient may complain of hoarseness.
4. Hypothyroidism affect women five times more frequently than men and occurs most often between 30 and 60 years of age
5. Severe hypothyroidism results in a subnormal temperature and pulse rate
6. The patient usually begins to gain weight even without an increased food intake
7. The hair thins and falls out.
8. The patient often complains of being cold even in a warm environment.



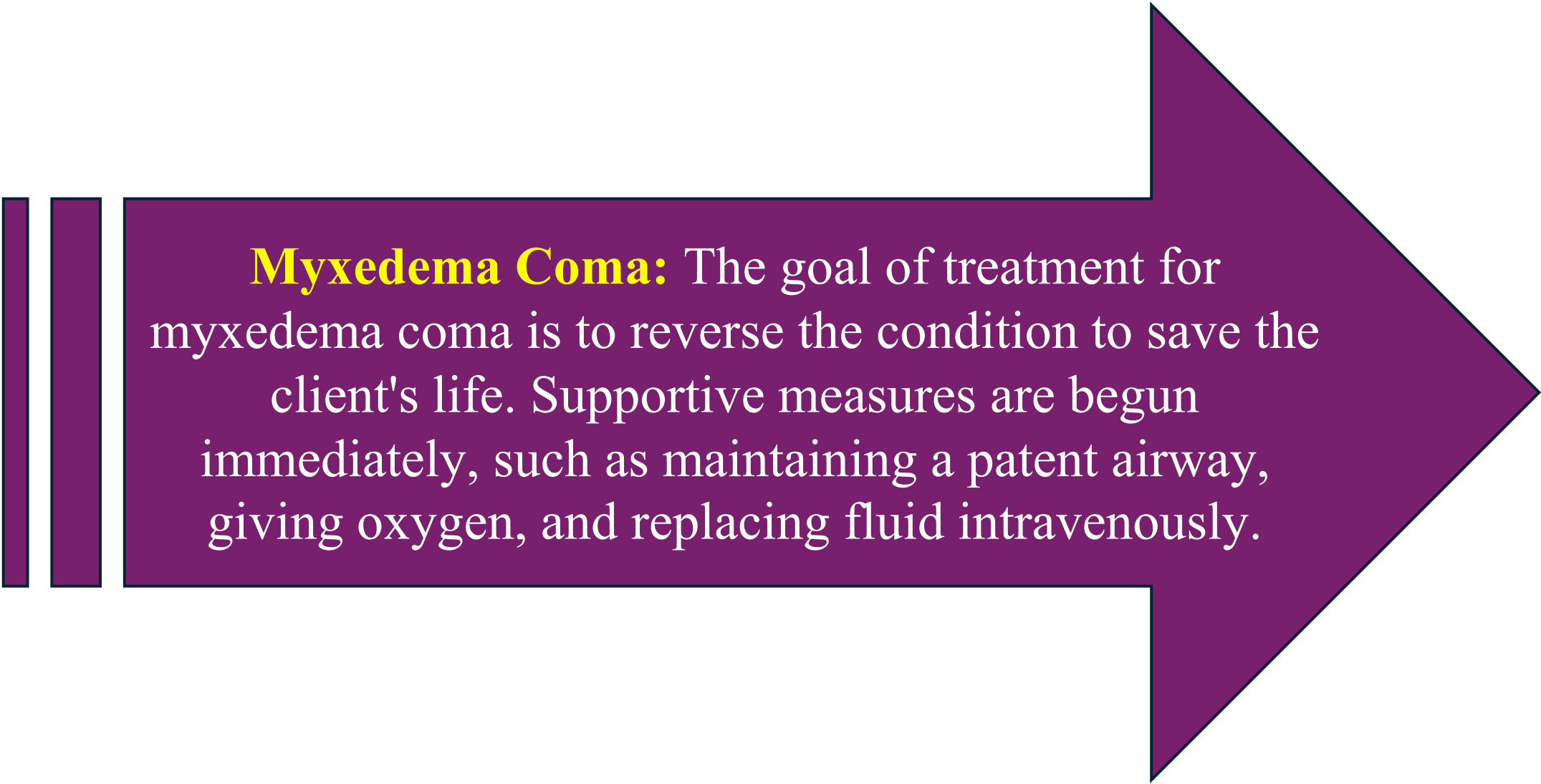
Medical management of hypothyroidism

Pharmacologic Management

Hypothyroidism. The goals of treatment for hypothyroidism are to correct thyroid hormone deficiency and reverse symptoms and prevent further cardiac and arterial damage.

For hypothyroidism to be reversed permanently, the client usually needs to take thyroid hormone preparation throughout life. Available thyroid medications include sodium levothyroxine, and desiccated thyroid, converted in the body to both T₄ and T₃

Clients with cardiac complications must be started on small doses of thyroid hormone. Large doses could precipitate heart failure or myocardial infarction



Myxedema Coma: The goal of treatment for myxedema coma is to reverse the condition to save the client's life. Supportive measures are begun immediately, such as maintaining a patent airway, giving oxygen, and replacing fluid intravenously.

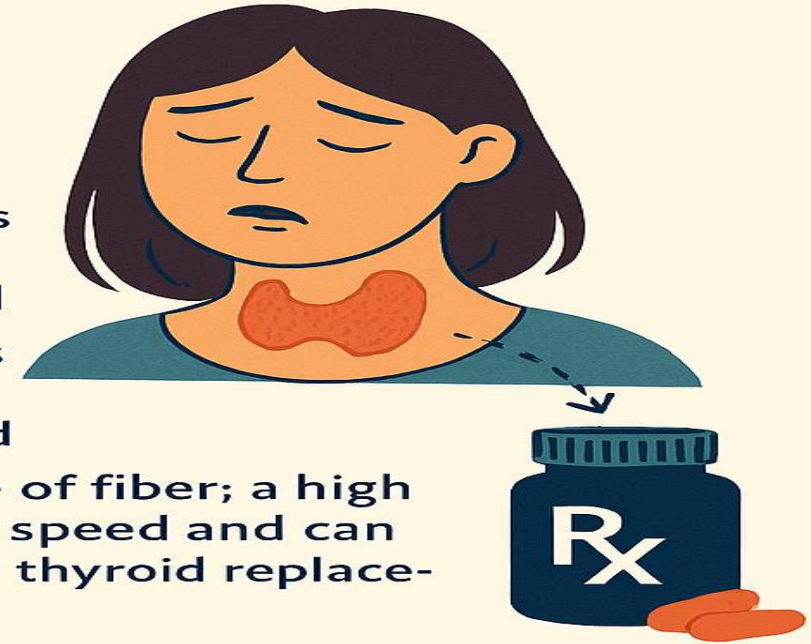
DIETARY MANAGEMENT

If the hypothyroidism and goiter are due to iodine deficiency the client should be on a diet higher in iodine.

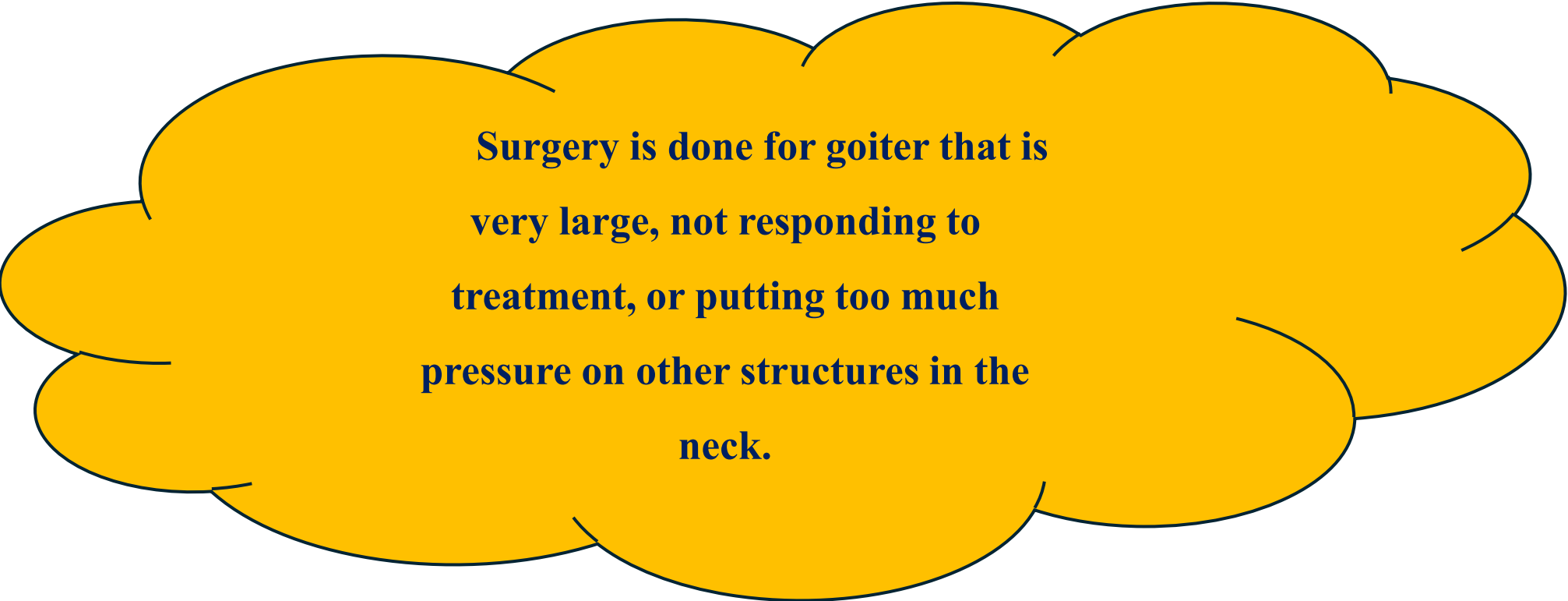
- Avoid excessive intake of these foods, which can inhibit thyroid utilization:
 - Cruciferous vegetables: broccoli, brussels sprouts, cabbage
 - Soybeans
 - Soy products
 - Spinach
 - Strawberries



- Note: Some research shows that walnuts may decrease the absorption of Synthroid
- Note: Some research shows that walnuts may decrease the absorption of Synthroid
- Aim for a consistent intake of fiber; a high fiber diet affects digestion speed and can reduce the ability to absorb thyroid replacement medication
- Allow a minimum of 2 hours separation between taking your thyroid medication and any supplements that contain iron or calcium
- Take thyroid medication on an empty stomach for optimal utilization



Surgical Management



Surgery is done for goiter that is very large, not responding to treatment, or putting too much pressure on other structures in the neck.

Nursing Management for hypothyroidism

ASSESSMENT

The nurse should carefully assess the client for the presence of signs and symptoms of hypothyroidism.



Nursing Diagnosis: Skin Integrity, High Risk for Impaired related to edema and dryness secondary to infiltration of fluid into interstitial spaces.

Planning: Expected Outcomes; The client will have all skin remain intact, as evidenced by absence of injury and resolution of edema.

Implementation: The nurse must monitor the sacrum coccyx, elbows, scapula, and other pressure points for signs of redness or tissue breakdown. The nurse must remember that edematous tissues are more prone to decubitus ulcer formation. The client should be placed on a strict turning schedule and on a pressure reduction mattress.

Nursing Diagnosis: Hypothermia related to slowed metabolic rate.

Planning: Expected Outcomes; The client will not develop hypothermia, as evidenced by a normal body temperature.

Implementation: Provide the client with a comfortable, warm environment. Remember that hypothyroidism sharply increases sensitivity to cold. If necessary, supply extra clothing and warm blankets.

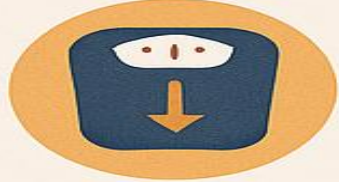
Nursing Diagnosis: Social Isolation related to lethargy, weakness, apathy, and changes in appearance.

Planning: Expected Outcomes; The client will return to a pre-illness level of social interaction, as evidenced by increasing interaction with others and statements of acceptance of reversibility of appearance changes.

Implementation: The nurse must reassure' the client and significant others that the clients appearance and energy level will gradually improve with thyroid hormone therapy.



HYPERTHYROIDISM



Weight
loss



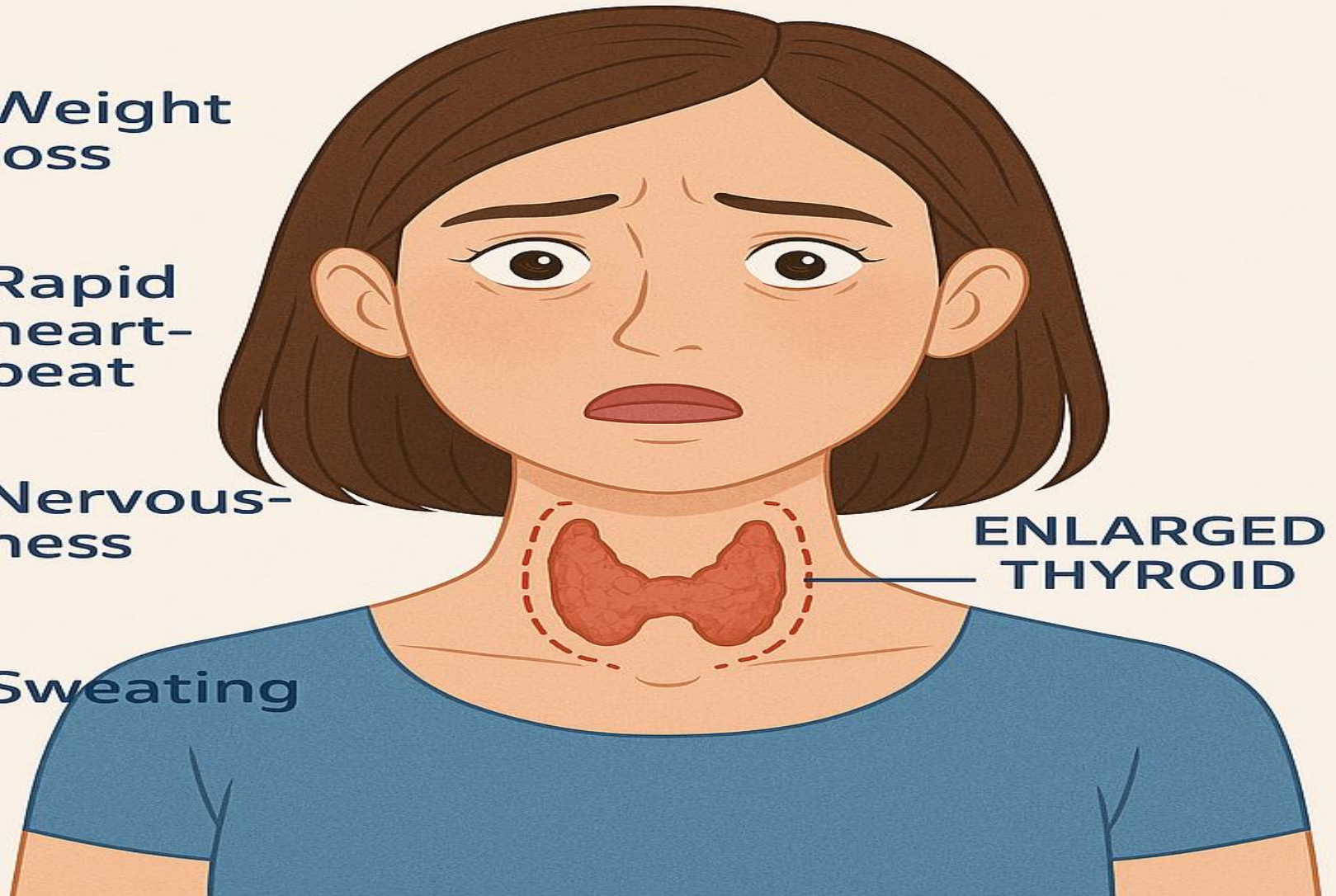
Rapid
heart-
beat

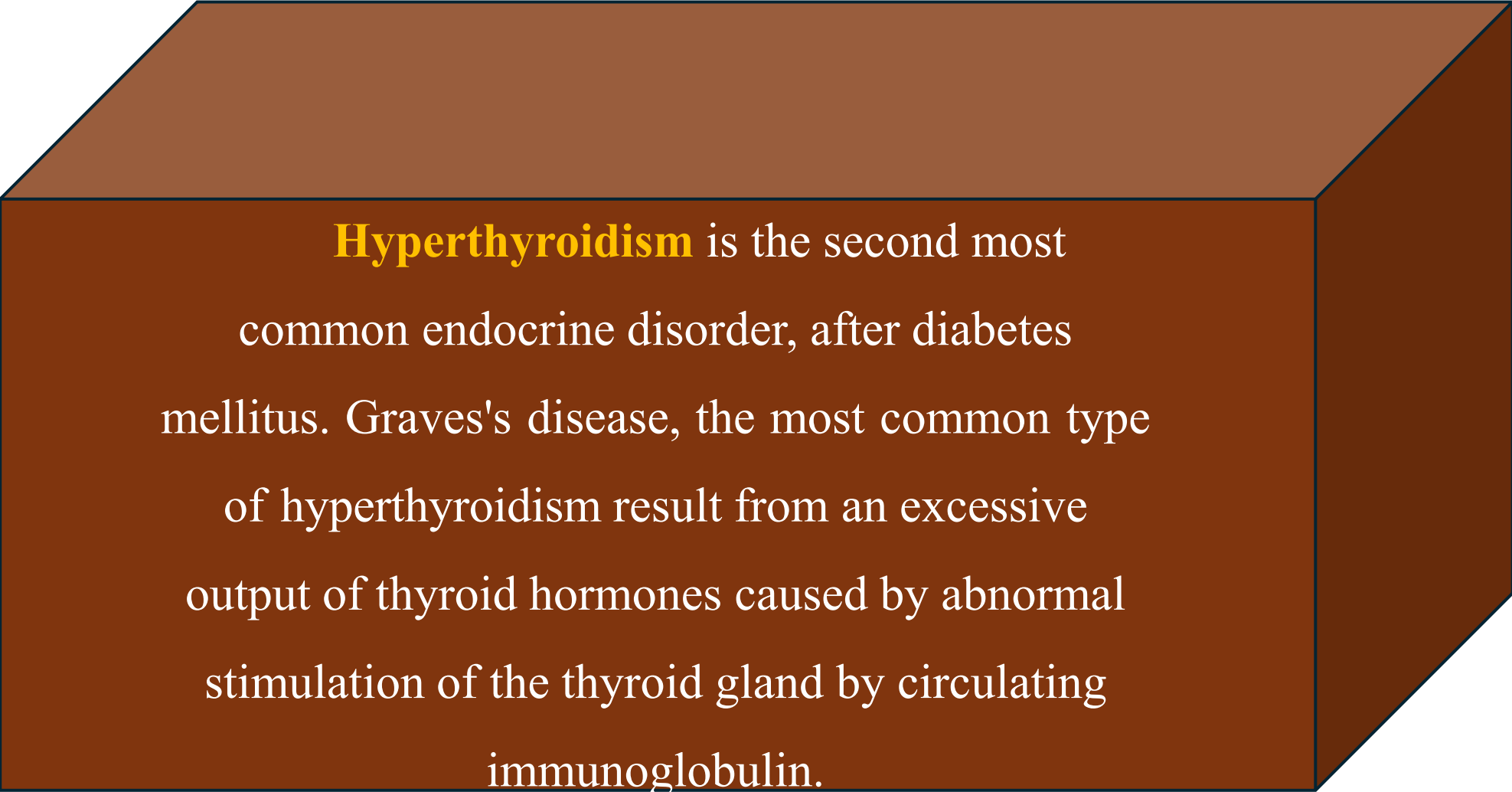


Nervous-
ness



Sweating

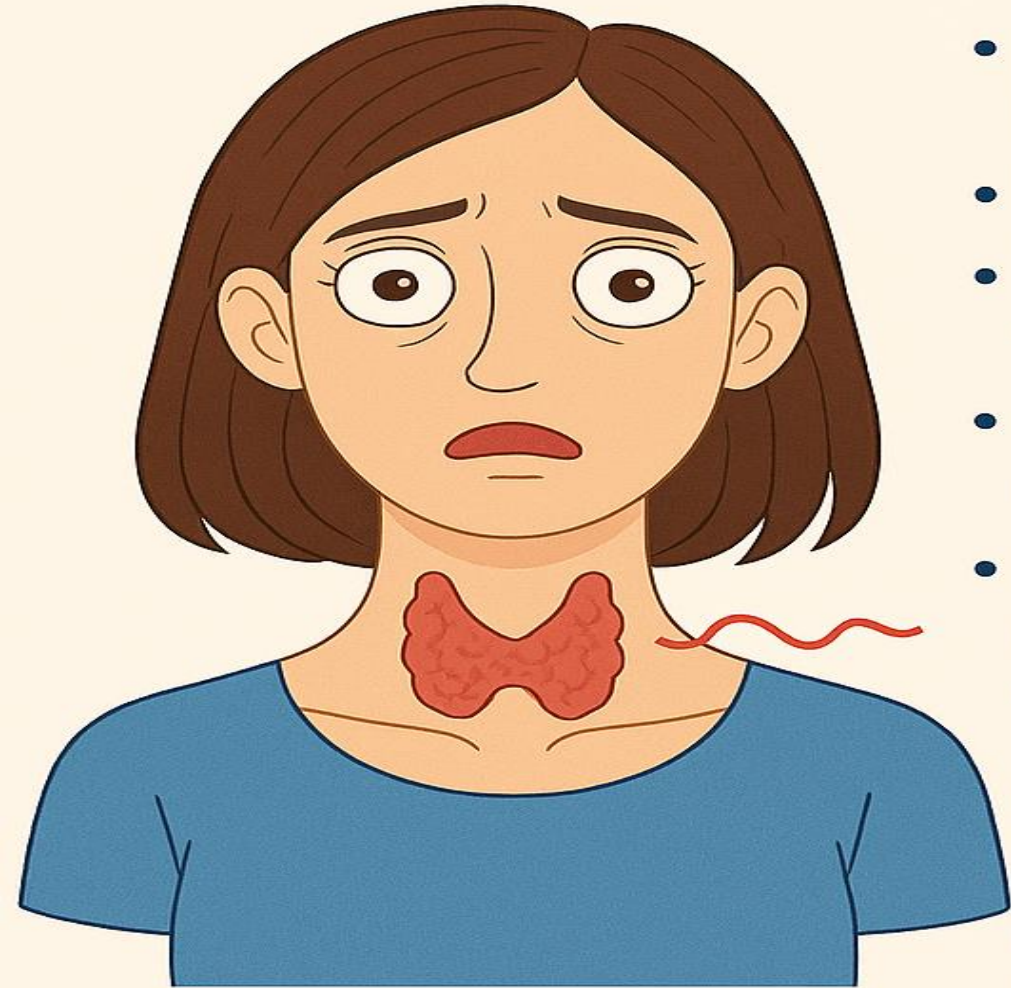




Hyperthyroidism is the second most common endocrine disorder, after diabetes mellitus. Graves's disease, the most common type of hyperthyroidism result from an excessive output of thyroid hormones caused by abnormal stimulation of the thyroid gland by circulating immunoglobulin.

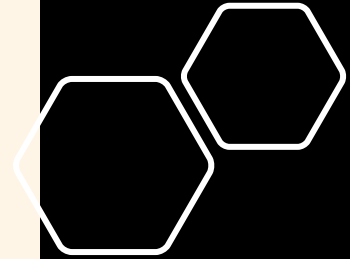
Hyperthyroidism affects women eight times than men. It may appear after emotional shock, stress or infection. Other common causes of hyperthyroidism include thyroiditis and excessive ingestion of thyroid hormone.

Clinical Manifestations



Pulse rate ranges between 90 and 160 beat / min.

- Nervousness irritability
 - Palpitations
 - They tolerate it poorly; the skin is flushed
 - Exophthalmoses (bulging eyes)
 - Increased appetite and dietary intake, progress weight loss
- Abnormal muscular fatigability and weakness
Amenorrhea and changes in bowel function
Osteoporosis and fracture



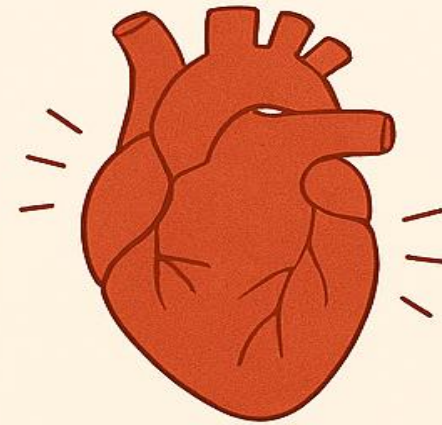
Diagnostic Assessment:

- The client's physical appearance (enlarged neck, protruding eyes, agitated expression; the symptoms of agitation, restlessness. and weight loss.
- Laboratory findings. The serum thyroid hormone levels are usually elevated; serum cholesterol levels are usually depressed.

COMPLICATIONS OF Graves' Disease



Exophthalmos



Heart disease



**Thyroid storm
(thyroid crisis,
thyrotoxicosis)**

Pulse rate ranges between 90 and 160 beat / min.

Medical Management of Hyperthyroidism

A. Pharmacologic Management:

1) Antithyroid medication



- Propanolol (β -blocker)



2) Radioiodine (capsule of radioactive iodine)

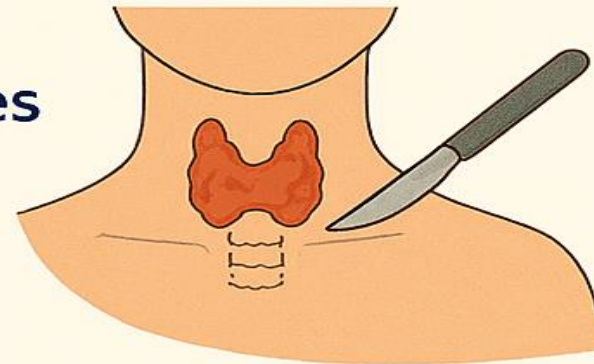
3) Thyroid storm (thyroid crisis, thyrotoxicosis)

B. Dietary Management:

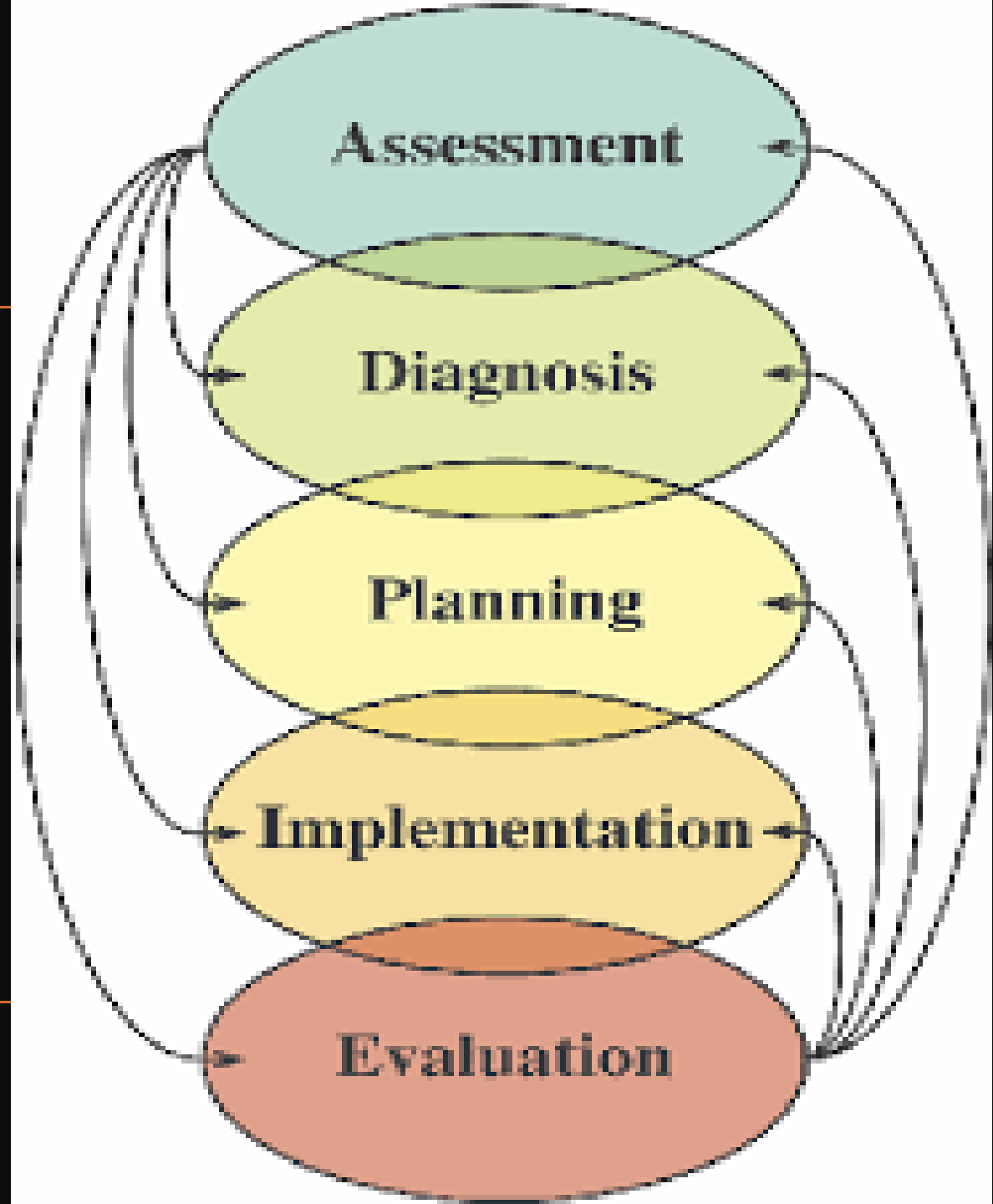
A diet of 4000 to 5000 calories with high protein levels

C. Surgery (thyroidectomy)

Removal of thyroid gland

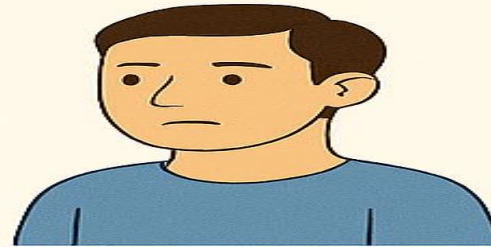
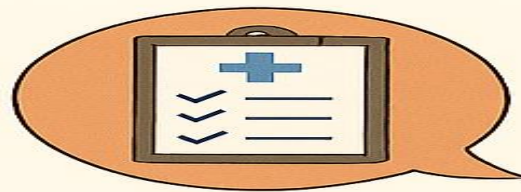


Nursing Process



ASSESSMENT

- The nurse should begin the assessment by obtaining a complete client history. By asking questions concerning weight, appetite, activity, heat intolerance, and bowel activity, the nurse can assess for the presence of typical signs and symptoms of hyperthyroidism.
- The nurse should assess the client for all the typical clinical manifestations associated with Graves' disease. The client should also be questioned regarding visual difficulties, fatigue, weakness, tremors, or insomnia.



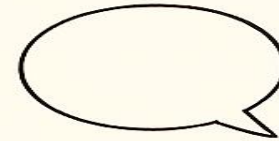
The nurse should also question significant others about mood alterations experienced by the client.

NURSING DIAGNOSIS

1. Nursing Diagnosis: Nutrition, altered:
Less than body requirements related to
accelerated metabolic rate.

Planning: Expected Outcomes:

The client's weight loss will stop, as evidenced
by client's ability to take in sufficient calories
to return to ideal body weight.



IMPLEMENTATION



- Provide the client with a well-balanced, high-calorie diet. Six full meals a day may be needed to satisfy their appetite. Encourage patient to eat foods that are nutritious and contain simple amounts of protein, carbohydrates, fats, and minerals. Clients should be weighed daily,

NURSING DIAGNOSIS

1. Nursing Diagnosis: Activity intolerance related to exhaustion.

Planning: Expected Outcomes:
The client will be able to engage in normal level of activity.

IMPLEMENTATION

- Provide the client with an environment that is restful both mentally and physically.



- Provide the client with an environment that is restful both mentally and physically. Assign these clients to a private room to promote rest and to prevent them from disturbing others through hyperactivity and restlessness.

COLLABORATIVE PROBLEM

1. Injury, high risk for; corneal ulcerations, infection, and possible blindness related to inability to close eyelids secondary to exophthalmos.

Planning: Expected Outcomes:
The nurse will prevent corneal ulceration, infection, or blindness, as evidenced by no further development of exophthalmos.

IMPLEMENTATION

- The treatment should be started as soon as possible once the diagnosis is made so exophthalmos can be avoided.
- The client should use artificial tears and eye patches as needed to prevent irritation.

